

| Question |     |      | Marking details                                                                                                                                                                                                                    | Marks available |            |     |       |       |      |
|----------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-----|-------|-------|------|
|          |     |      |                                                                                                                                                                                                                                    | AO1             | AO2        | AO3 | Total | Maths | Prac |
| 4        | (a) |      | Reasonable attempt at sinusoidal curve (1)<br>Accurate curve drawn with maximum and minimum at correct points (1)<br>N.B. Look out for points being missed, trough in the right place, starting too low                            |                 | 2          |     | 2     | 1     | 2    |
|          | (b) |      | Period – the time taken for one complete cycle of the oscillations (1)<br>$0.9 \pm 0.01$ [s] (1)                                                                                                                                   | 1               | 1          |     | 2     |       | 2    |
|          | (c) |      | $y = 1.5 \cos\left(\frac{2\pi}{0.9[1]}t \pm \pi(1)\right)$ cm <b>ecf</b> on period<br>Don't accept 180                                                                                                                             |                 | 2          |     | 2     | 1     | 1    |
|          | (d) | (i)  | Good tangent (1)<br>Substitution of values for the gradient <b>ecf</b> (1)<br>Max velocity i.e. $10.5 \pm 1.0$ [cm s <sup>-1</sup> ] (1) <b>ecf</b>                                                                                | 1<br>1          | 1          |     | 3     | 2     | 3    |
|          |     | (ii) | Max velocity of 10 [cm s <sup>-1</sup> ] <b>ecf</b> (1)<br>[Sinusoidal curve with] appropriate period <b>ecf</b> (1)<br>Appropriate phase for starting point i.e. sin curve (1)                                                    |                 | 3          |     | 3     | 2     | 3    |
|          | (e) | (i)  | Acceleration is [directly] proportional to displacement (1)<br>And is directed towards a fixed point (1)                                                                                                                           | 2               |            |     | 2     |       |      |
|          |     | (ii) | Substitution: $T = 2\pi\sqrt{\frac{m}{k}}$ i.e. $0.9 = 2\pi\sqrt{\frac{0.030}{k}}$ (1) <b>ecf</b><br><br>Rearrangement: $k = \frac{4\pi^2 \times 0.030}{0.81}$ (1)<br><br>Answer = 1.46 N m <sup>-1</sup> <b>**unit mark**</b> (1) | 1               | 1<br><br>1 |     | 3     | 2     | 3    |